



# IndianOil

## JOURNEY RISK MANAGEMENT (JRM) STUDY

### Salem Terminal to SRI UDAYAGIRI FUELS KSK

#### Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

#### Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

#### Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

## **Compliance with the Road Transport Safety Policy (RTSP)**

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

## **Emergency Preparedness and Response**

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

## **Environmental Considerations**

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

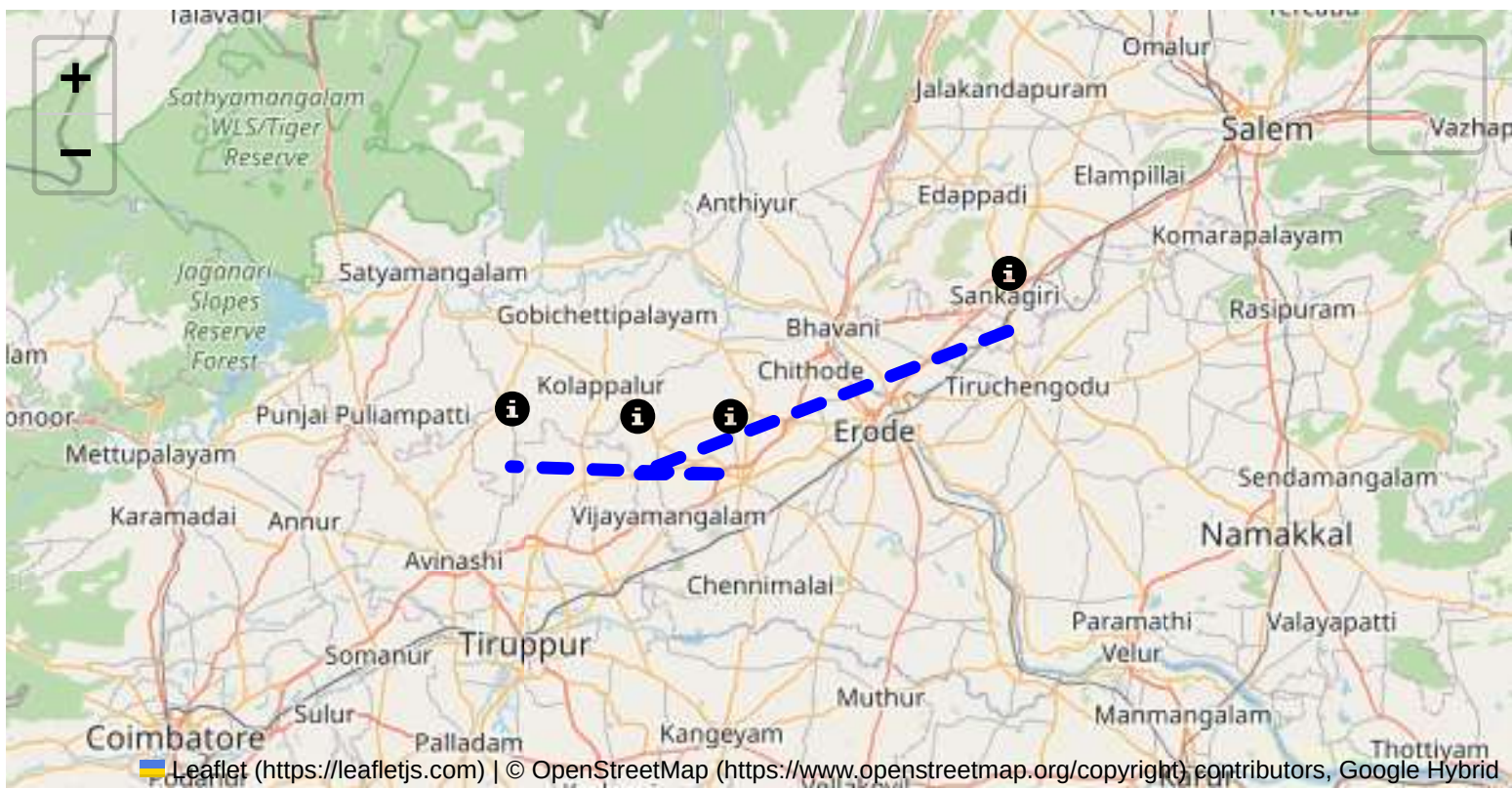
## **Journey Risk Mitigation**

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

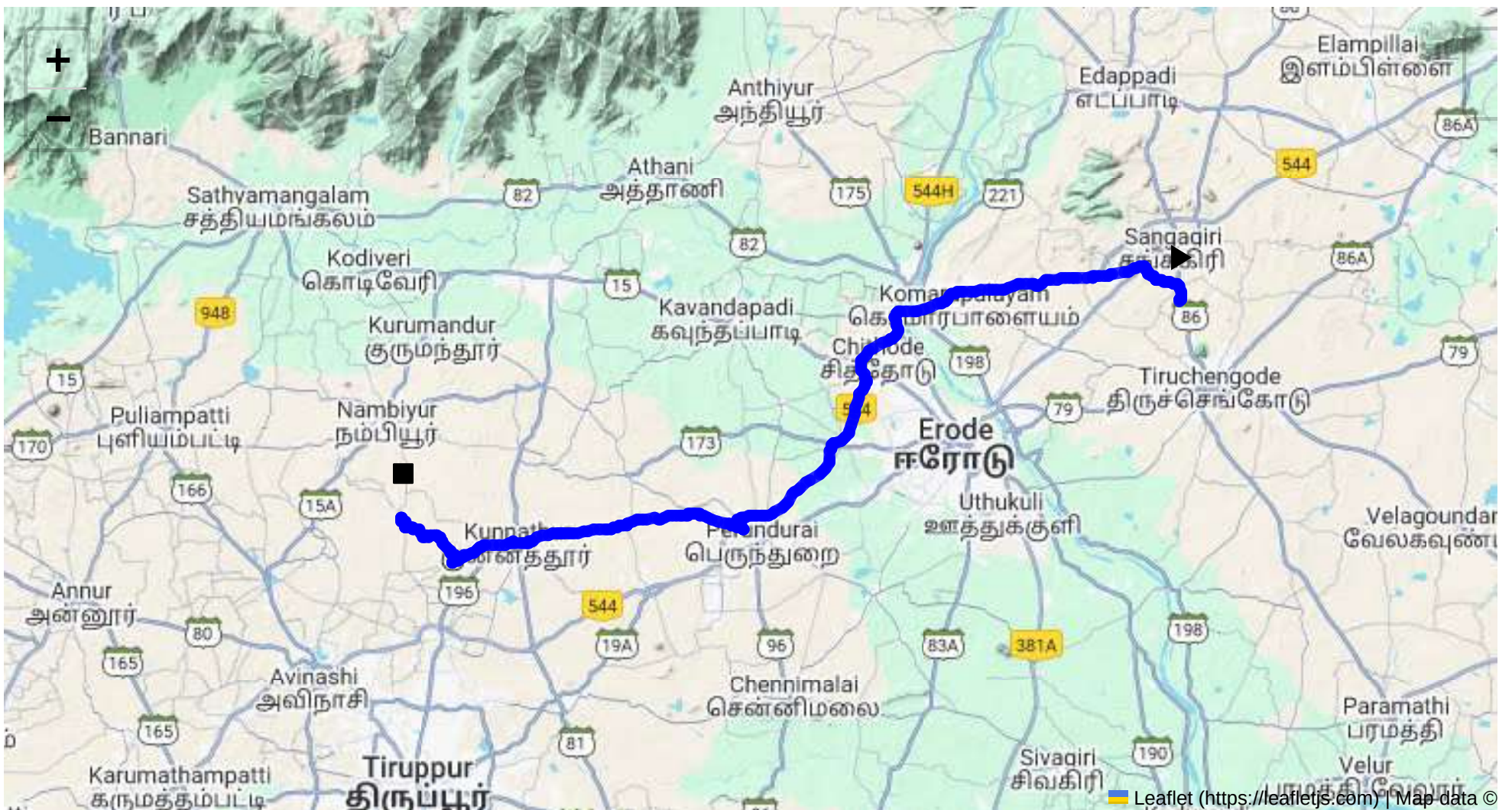
## **Compliance with Government Directives**

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.





**Route Summary:**  
**Total Distance: 78.35 km**  
**Estimated Duration: 1.7 hours**  
**Adjusted Duration (Heavy Vehicle): 2.2 hours**  
**Start: (11.4381, 77.8734)**  
**End: (11.290265, 77.329872)**



## Welcome to the Journey Risk Management Study

### Route Analysis Report

#### 1. Overview of the Route Map

The route begins at Sangagiri in Tamil Nadu, passing through SH 196 and Perundurai, and concludes at Olalakovil via Avinashi-Palangarai-Nambiyur Road. It stretches approximately 78.35 kilometers and is



predominantly rural with some small town crossings.

## **2. Typical Weather Conditions and Potential Weather-Related Hazards**

Tamil Nadu experiences tropical weather with hot summers, a monsoon season, and mild winters. Monsoon months (June to September) could bring heavy rains leading to waterlogging or reduced visibility. Occasional thunderstorms are also a risk which can impact driving conditions, particularly on rural roads.

## **3. Analysis of Traffic Patterns**

Traffic is typically moderate but can become congested during the early morning and late afternoon hours as people commute for work and school. Perundurai is a potential congestion point as it is a notable town along the route, especially during weekends or local market days.

## **4. Assessment of Road Quality and Infrastructure**

Road conditions along SH 196 are generally good, but variations in quality exist. Potholes and uneven road surfaces can be found, particularly in rural stretches. The infrastructure improves near town centers but lessens in more isolated areas.

## **5. Suggestions for Alternative Routes for Emergencies**

In case of road closure or major delays, an alternative route could involve diverting through the NH 544 (Salem-Kochi-Kanyakumari Highway) which is approximately parallel but may be longer in distance.

## **6. Summary of Local Regulations Affecting Hazardous Material Transport**

Transporting hazardous materials requires compliance with Tamil Nadu's strict protocols, including possessing proper documentation and permits. There are designated times when heavy vehicles are restricted from accessing certain urban areas.

## **7. Overview of Historical Incidents Involving Heavy Vehicles**

While there are periodic incidents involving heavy vehicles, including rollovers or collisions mainly due to speeding or overloaded vehicles, there are no major recurring incidents involving hazardous material reported specifically along this route.

## **8. Environmental Considerations and Sensitive Areas**

There are a few environmentally sensitive areas along the route, including agricultural lands and water bodies that require cautious driving to prevent contamination or disruption.

## **9. Analysis of Communication Coverage**

Mobile signal coverage is generally consistent along the main highways but can become patchy in rural or forested areas. Dead zones are most likely in isolated stretches between towns.

## 10. Estimated Emergency Response Times

Emergency services in urban areas like Perundurai tend to respond quickly (within 15-20 minutes), whereas in rural areas, response times could extend to 30-45 minutes due to longer travel distances.

## 11. Overall Summary of Risk Assessment

Overall, the risk on this route is moderate. Key risks include potential weather impacts during the monsoon, occasional traffic congestion, and variable road conditions. Due to these factors and the potential for communication signal loss in certain areas, preparedness in the form of route planning and compliance with local transport regulations is crucial for safe transportation of hazardous materials.

**Recommendation:** Drivers should be updated about weather conditions and advised to plan travel avoiding peak traffic hours whenever possible. It is imperative to have all required permits for hazardous material transport and have emergency contacts readily available. Regular maintenance checks on vehicles can help prevent accidents related to mechanical failures.

### Risk Assessment - Turns

|    | Risk Type  | Risk Level | Coordinates        | Speed Limit |
|----|------------|------------|--------------------|-------------|
| 0  | Turn       | High       | 11.43968, 77.87345 | 15 KM/Hr    |
| 1  | Turn       | High       | 11.44024, 77.87528 | 15 KM/Hr    |
| 2  | Turn       | High       | 11.44898, 77.87410 | 15 KM/Hr    |
| 3  | Turn       | Medium     | 11.45357, 77.85789 | 30 KM/Hr    |
| 4  | Turn       | High       | 11.45352, 77.85698 | 15 KM/Hr    |
| 5  | Turn       | Medium     | 11.45847, 77.85140 | 30 KM/Hr    |
| 6  | Turn       | Medium     | 11.45898, 77.85135 | 30 KM/Hr    |
| 7  | Turn       | High       | 11.46303, 77.84945 | 15 KM/Hr    |
| 8  | Turn       | Medium     | 11.45509, 77.81383 | 30 KM/Hr    |
| 9  | Turn       | Medium     | 11.45085, 77.77020 | 30 KM/Hr    |
| 10 | Turn       | Medium     | 11.43165, 77.67734 | 30 KM/Hr    |
| 11 | Blind Spot | Blind Spot | 11.28888, 77.56550 | 10 KM/Hr    |
| 12 | Blind Spot | Blind Spot | 11.28189, 77.56929 | 10 KM/Hr    |
| 13 | Turn       | Medium     | 11.29281, 77.53814 | 30 KM/Hr    |
| 14 | Turn       | High       | 11.29251, 77.53037 | 15 KM/Hr    |

|    | Risk Type  | Risk Level | Coordinates        | Speed Limit |
|----|------------|------------|--------------------|-------------|
| 15 | Turn       | Medium     | 11.29192, 77.52990 | 30 KM/Hr    |
| 16 | Turn       | Medium     | 11.27393, 77.42230 | 30 KM/Hr    |
| 17 | Turn       | Medium     | 11.27378, 77.41973 | 30 KM/Hr    |
| 18 | Turn       | Medium     | 11.27288, 77.41914 | 30 KM/Hr    |
| 19 | Turn       | Medium     | 11.27254, 77.41851 | 30 KM/Hr    |
| 20 | Turn       | Medium     | 11.27228, 77.41323 | 30 KM/Hr    |
| 21 | Turn       | Medium     | 11.27353, 77.40862 | 30 KM/Hr    |
| 22 | Turn       | High       | 11.27057, 77.38574 | 15 KM/Hr    |
| 23 | Turn       | Medium     | 11.26001, 77.36845 | 30 KM/Hr    |
| 24 | Turn       | Medium     | 11.25977, 77.36633 | 30 KM/Hr    |
| 25 | Blind Spot | Blind Spot | 11.25985, 77.36605 | 10 KM/Hr    |
| 26 | Turn       | Medium     | 11.26055, 77.36646 | 30 KM/Hr    |
| 27 | Turn       | Medium     | 11.26413, 77.36636 | 30 KM/Hr    |
| 28 | Turn       | Medium     | 11.26461, 77.36664 | 30 KM/Hr    |
| 29 | Blind Spot | Blind Spot | 11.26487, 77.36723 | 10 KM/Hr    |
| 30 | Turn       | High       | 11.26516, 77.36707 | 15 KM/Hr    |
| 31 | Turn       | High       | 11.26513, 77.36557 | 15 KM/Hr    |
| 32 | Turn       | Medium     | 11.26796, 77.36396 | 30 KM/Hr    |
| 33 | Turn       | High       | 11.27740, 77.35788 | 15 KM/Hr    |
| 34 | Blind Spot | Blind Spot | 11.27682, 77.34555 | 10 KM/Hr    |
| 35 | Turn       | Medium     | 11.28158, 77.34302 | 30 KM/Hr    |
| 36 | Turn       | Medium     | 11.28239, 77.33843 | 30 KM/Hr    |
| 37 | Turn       | Medium     | 11.28317, 77.33761 | 30 KM/Hr    |
| 38 | Turn       | High       | 11.28349, 77.33752 | 15 KM/Hr    |
| 39 | Blind Spot | Blind Spot | 11.28369, 77.33267 | 10 KM/Hr    |
| 40 | Turn       | Medium     | 11.28760, 77.33292 | 30 KM/Hr    |
| 41 | Turn       | Medium     | 11.28784, 77.33263 | 30 KM/Hr    |
| 42 | Turn       | High       | 11.28799, 77.32982 | 15 KM/Hr    |
| 43 | Turn       | High       | 11.29047, 77.32961 | 15 KM/Hr    |

## Emergency Locations

|    | type     | name                                           | coordinates               | speed_limit | risk_level |
|----|----------|------------------------------------------------|---------------------------|-------------|------------|
| 3  | hospital | Dhanvantri Multi Speciality Hospital           | 11.451362, 77.766602      | 30 km/h     | Medium     |
| 4  | hospital | Dhanvanthri Hospital                           | 11.4496712,<br>77.7593772 | 30 km/h     | Medium     |
| 5  | hospital | J.K.K. Trust Hospital                          | 11.4445841,<br>77.7307962 | 30 km/h     | Medium     |
| 7  | hospital | Shri Sathyanarayana Hospital                   | 11.4291297,<br>77.6913408 | 30 km/h     | Medium     |
| 8  | hospital | Thanish Siddha Hospital                        | 11.430003, 77.674964      | 30 km/h     | Medium     |
| 9  | clinic   | Harshitha Clinic                               | 11.4313207,<br>77.674718  | 30 km/h     | Medium     |
| 10 | hospital | Sri Kaalangi Siddhar Mooligai Vaithiya Nilayam | 11.432369, 77.674894      | 30 km/h     | Medium     |
| 11 | clinic   | G.K Clinic                                     | 11.4297244,<br>77.6749715 | 30 km/h     | Medium     |
| 12 | clinic   | Erode Cancer Centre                            | 11.3732, 77.649152        | 30 km/h     | Medium     |
| 14 | hospital | Irt Hospital                                   | 11.2803603,<br>77.5644118 | 30 km/h     | Medium     |
| 15 | hospital | IRT Ramalingam Tuberculosis Hospital           | 11.2828419,<br>77.5628621 | 30 km/h     | Medium     |
| 16 | hospital | Sanitorium Hospital                            | 11.2828041,<br>77.5629626 | 30 km/h     | Medium     |
| 17 | clinic   | Sumathi Clinic                                 | 11.2725128,<br>77.411116  | 30 km/h     | Medium     |
| 18 | hospital | Thavamani Hospital                             | 11.2719549,<br>77.4120852 | 30 km/h     | Medium     |
| 19 | hospital | Viswanathan Hospital                           | 11.2725112,<br>77.411171  | 30 km/h     | Medium     |
| 20 | clinic   | Harisandhara Clinic                            | 11.2711263,<br>77.4109069 | 30 km/h     | Medium     |
| 21 | hospital | S V Hospital                                   | 11.273427, 77.411431      | 30 km/h     | Medium     |
| 22 | hospital | Dhanam Thangam Hospital                        | 11.2719384,<br>77.4121472 | 30 km/h     | Medium     |
| 23 | police   | Kunnathur Police Station                       | 11.2730826,<br>77.4111808 | 30 km/h     | Medium     |
| 24 | clinic   | Arogya Clinic                                  | 11.2706519,<br>77.4100587 | 30 km/h     | Medium     |
| 25 | clinic   | Sri Murugan Nursing Home                       | 11.271798, 77.409776      | 30 km/h     | Medium     |



## Crowded Spots

|    | type        | name                                      | coordinates               | speed_limit | risk_level |
|----|-------------|-------------------------------------------|---------------------------|-------------|------------|
| 0  | school      | KRP Matric. Hr. Sec School                | 11.4546193,<br>77.8142445 | 30 km/h     | Medium     |
| 1  | college     | Vivekanandha Engineering College          | 11.4589312,<br>77.7899284 | 30 km/h     | Medium     |
| 2  | marketplace | Monday market                             | 11.452863,<br>77.775989   | 30 km/h     | Medium     |
| 6  | school      | SSM Matriculation Higher Secondary School | 11.4321653,<br>77.6880046 | 30 km/h     | Medium     |
| 13 | college     | Government polytechnic college            | 11.2907053,<br>77.5698726 | 30 km/h     | Medium     |

## Route Photos of Risky Spots



**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.44024, 77.87528





**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.44898, 77.87410



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.45357, 77.85789





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.45352, 77.85698



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.45847, 77.85140



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.45898, 77.85135





**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.46303, 77.84945



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.45509, 77.81383





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.45085, 77.77020



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.43165, 77.67734



**Risk Type:** Blind Spot  
**Risk Level:** Blind Spot  
**Speed Limit:** 10 KM/Hr  
**Coordinates:** 11.28888, 77.56550





**Risk Type:** Blind Spot  
**Risk Level:** Blind Spot  
**Speed Limit:** 10 KM/Hr  
**Coordinates:** 11.28189, 77.56929



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.29281, 77.53814





**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.29251, 77.53037



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.29192, 77.52990



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.27393, 77.42230





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.27378, 77.41973



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.27288, 77.41914





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.27254, 77.41851



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.27228, 77.41323



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.27353, 77.40862





**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.27057, 77.38574



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.26001, 77.36845





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.25977, 77.36633



**Risk Type:** Blind Spot  
**Risk Level:** Blind Spot  
**Speed Limit:** 10 KM/Hr  
**Coordinates:** 11.25985, 77.36605



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.26055, 77.36646





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.26413, 77.36636



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.26461, 77.36664





**Risk Type:** Blind Spot  
**Risk Level:** Blind Spot  
**Speed Limit:** 10 KM/Hr  
**Coordinates:** 11.26487, 77.36723



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.26516, 77.36707



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.26513, 77.36557





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.26796, 77.36396



**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.27740, 77.35788





**Risk Type:** Blind Spot  
**Risk Level:** Blind Spot  
**Speed Limit:** 10 KM/Hr  
**Coordinates:** 11.27682, 77.34555



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.28158, 77.34302



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.28239, 77.33843





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.28317, 77.33761



**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.28349, 77.33752





**Risk Type:** Blind Spot  
**Risk Level:** Blind Spot  
**Speed Limit:** 10 KM/Hr  
**Coordinates:** 11.28369, 77.33267



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.28760, 77.33292



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.28784, 77.33263





**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.28799, 77.32982



**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.29047, 77.32961

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